

Entropy Collapse Has a Direction: Correcting the Visitation Channel in Token-Level Entropy Control for RLVR

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Abstract

Reinforcement learning with verifiable rewards (RLVR) reliably drives a policy’s entropy toward zero, and a growing family of methods intervenes by re-weighting the policy loss with a per-token estimate of how much each token’s update costs in entropy. We show that every such method estimates only half of the relevant quantity. The gradient of the *trajectory* entropy splits exactly into two terms: a *local* channel, which asks how the conditional distribution at a visited state flattens or sharpens, and a *visitation* channel, which asks how the probability of *reaching* that state changes. Existing token-level estimators capture the first and are structurally blind to the second. The blindness is worse than an omission: the local estimator is exactly zero at a uniform branch point, that zero is the correct first-order answer, and the visitation term’s prefactor is simultaneously maximal there — the two channels are *anti-correlated*, so no refinement of the local term can recover the missing signal. We derive the missing channel into a per-token advantage A_H whose reward-to-go is the remaining conditional entropy H_{togo} . Because RLVR updates only after a rollout is complete, H_{togo} can be read off the entropies the policy actually had along the sampled continuation, so the method, STEER-V, needs no auxiliary model; a learned multi-token forecaster of the same quantity is evaluated as an ablation. The sibling-prefix baseline that keeps the estimator unbiased is supplied by a tree-structured rollout, and because A_H sums to zero within a sibling set, the correction cannot change the average amount of damping — only which of two siblings is damped. We test these consequences with LoRA fine-tuning of a 1.5B mathematics model in a six-arm design that runs the full pipeline with A_H ’s value destroyed in two different ways and with the forecaster in its place, at three seeds, under an analysis fixed before the first run. Switching the damping apparatus moves the aggregate entropy by $-$ and $-$, and

changing only what A_H carries moves it by $-$ — a separation the registered test calls $-$. On six benchmarks at the final checkpoint, STEER-V leads the value-destroyed control on $-$ of six, which the pre-registered thresholds call $-$.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model’s output distribution. Measured as the mean per-token entropy of the policy, this sharpening is fast, monotone, and, past a point, coincident with the end of useful learning. The standard reading is that the collapse is the problem, and the standard remedy is to spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose the most entropy, or — in the most recent line of work — a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token’s update destroys.

We argue that the accounting behind that remedy is incomplete, and that the incompleteness is structural rather than a matter of estimator quality.

Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution the policy induces over whole responses to a prompt x . The autoregressive factorization turns it into an expectation of a sum of per-token conditional entropies, and the gradient of that expression splits, by the product rule, into exactly two terms. The first holds the visited states fixed and asks how the conditional distribution at each one changes: this is what a per-token entropy estimate measures. The second holds the per-state entropies fixed and asks how the probability of *visiting* each state changes. An update that concentrates mass on one continuation makes the states downstream of that continuation more likely and the states downstream of its abandoned siblings less likely; the diversity those siblings carried leaves the trajectory distribution even though no conditional distribution anywhere became more peaked.

Our first contribution is to show that the local channel is not merely blind to the second term but *anti-correlated* with it. The local estimator factors into the advantage, a saturation factor, and the deviation of the sampled token’s surprisal from the mean surprisal — and that last factor has mean zero by the definition of entropy. At a branch point where several continuations are close to equiprobable it vanishes identically. The vanishing is not an estimator failure: a uniform distribution maximizes entropy, so the first-order change is genuinely zero and the damage is second order (Section 4.4). Meanwhile the visitation term’s prefactor is $1 - \pi_\theta(y_t | s_t)$, which is maximal precisely at those flat states. The local channel therefore spends its budget where the conditional entropy is fragile — rare tokens sampled from confident distributions — and spends nothing where the trajectory entropy is most at risk.

Our second contribution is an estimator for the missing channel that requires no additional rollouts and no auxiliary model. Eliminating the non-causal cross terms turns the visitation term into a REINFORCE estimator whose reward-to-go is the sum of the *remaining* conditional entropies, H_{togo} ; subtracting any function of the state leaves it unbiased. In RLVR the update is made only after the rollout has finished, and the trainer already evaluates the policy’s conditional entropy at every position of the sampled response, so H_{togo} — truncated and discounted — is available for free along the continuation the rollout actually took. We call the method that uses it STEER-V, for the visitation channel. The state-only baseline comes from rollouts that share a prefix. Plain group-relative rollouts almost never share prefixes, so we schedule the sharing: a tree-structured sampler branches a single root at fixed depths, and the resulting sibling sets give the baseline an actual support (Section 9). A parallel multi-token-prediction (MTP) forecaster of the same reward-to-go is the natural alternative, and we run it as an ablation rather than assume either estimator is the better one (Section 12.6).

The resulting per-token score A_H is a deviation from a sibling mean and therefore sums to zero over each sibling set. This has a consequence we state up front because it determines what evidence can count: the method *cannot* raise the average weight at branch points, only spread the weights of siblings apart. “Total entropy went up” is therefore not a prediction of this method, and an experiment looking for it is testing a different hypothesis (Sec-

tion 9.3).

The experiments fine-tune with LoRA (Hu et al., 2022). The reason is compute, and we say so rather than dress it as a design principle: it is what lets every arm run at three seeds, with its ablations, inside a fixed budget. Every arm — the baselines included — is trained with the identical adapter configuration, so the claims are comparisons between methods under LoRA; whether they carry over to full fine-tuning is not claimed here. The analysis, including what would count against us and a check of whether entropy collapses under LoRA at all, was committed before the first run (Appendix C).

We make the following contributions.

- An exact two-channel decomposition of the trajectory-entropy gradient, and an argument that the channel captured by current token-level entropy-control methods is anti-correlated with the channel they miss (Sections 4–4.4).
- A derivation of the missing channel into a per-token advantage A_H with no approximation up to the point of estimating H_{togo} , an estimator of it that reads the realised continuation and adds no parameters, and an explicit ledger of the approximations, the sign of each one’s error, and the mechanism that absorbs it (Sections 5–10).
- A tree-structured rollout that supplies the sibling-prefix baseline, and a bound showing what fraction of positions can carry a non-zero A_H under *any* sampler (Section 9).
- A six-arm experiment that separates the contribution of A_H ’s *value* from that of the apparatus delivering it, by running the full pipeline twice with the value destroyed in two different ways, and that tests the estimator by running the same method with a learned forecaster in its place (Sections 11 and 12).
- A test of the dissociation the construction predicts — that switching the damping apparatus moves the aggregate entropy while changing only what A_H carries does not — registered in advance as a comparison between groups of contrasts rather than read off any single one (Section 12.3).

2 Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented, and a range of interventions add an explicit entropy term to the objective or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization does not prevent the collapse and that continuing to train under it degrades accuracy (Anonymous, 2025), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Assumption. Both Hao et al. (2026) and this work rely on one structural assumption, which we restate so that our approximation ledger can refer to it.

Assumption 1 (parameter-independent softmax; Hao et al., 2026). For any state s , each action a carries an independent logit $z_{s,a}(\theta)$, so that a gradient step on the sampled token does not substantially move the logits of the other tokens at that state.

It is false for a network with shared parameters. We inherit it rather than introduce it, but our construction adds a second estimator that rests on it.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token’s update induces (Hao et al., 2026). That work observes that earlier interventions are “indirect” and that their effect is therefore limited and can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Wang et al. (2025d) show that restricting policy-gradient updates to a small minority of high-entropy “forking” tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis

predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Local stochasticity without global diversity. Wu et al. (2025a) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Estimating a reward-to-go of entropies. The quantity our estimator needs is a sum of future conditional entropies. We read it off the sampled continuation, which is what a REINFORCE estimator does with any reward-to-go. The alternative we compare against predicts it from the current hidden state with Medusa-style parallel prediction heads (Cai et al., 2024), used not for their intended purpose (speculative decoding) but as an entropy forecaster; Section 6 shows that forecast has a signed, monotone error.

Parameter-efficient RL fine-tuning. Low-rank adaptation (Hu et al., 2022) trains a small number of added parameters beside a frozen base model and is widely used to fine-tune language models with reinforcement learning under a limited budget. We use it for that reason. A low-rank update constrains how the policy can move, which may itself change how quickly it sharpens; we therefore measure whether entropy collapses under our configuration before reading anything else (Section 12.1).

3 Preliminaries

Notation. A prompt is x , a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot | s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta | x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a | s_t)$, so that $H = \mathbb{E}_{a \sim \pi_\theta}[I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \text{mean}_j r_j) / \text{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function

with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and instead multiply each token’s contribution to the clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from a first-order estimate of the entropy change that token’s update induces. Their Theorem 1 gives that estimate in closed form: under a parameter-independent softmax (their Assumption 1; our 2), the change in the conditional entropy at a *fixed* state s after one GRPO step is, to first order,

$$\Omega(s) = -\frac{\eta}{L} \mathbb{E}_a \left[\frac{\mathbb{I}_{\text{clip}}(s, a) A(s, a)}{\pi_{\text{old}}(a | s)} \cdot \pi_{\theta}(a | s) (1 - \pi_{\theta}(a | s)) \cdot (\log \pi_{\theta}(a | s) + H) \right], \quad (1)$$

where η is the learning rate, L the total decoded length in the update, and $\mathbb{I}_{\text{clip}}$ the PPO clipping indicator. The implementation evaluates the bracket at the sampled token rather than in expectation. The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, with

$$\alpha_u = \exp\left(-\alpha \frac{|\Omega_u|}{\max_v |\Omega_v|}\right), \quad \lambda_{\min} = e^{-\alpha}, \quad (2)$$

a batch-normalized exponential map into $[\lambda_{\min}, 1]$, so the weight can only attenuate. The factor $1/\pi_{\text{old}}$ in Eq. (1) will matter later: it makes Ω heavy-tailed, which is why the two channels must be normalized before they are combined (Section 8).

4 Two Channels

4.1 Step 1: the chain rule

The object we would like to protect is the entropy of the distribution the policy induces over *whole responses*,

$$\mathcal{H}(\pi_{\theta} | x) := - \sum_y \pi_{\theta}(y | x) \log \pi_{\theta}(y | x), \quad (3)$$

a sum over an exponentially large set. The autoregressive factorization $\pi_{\theta}(y | x) = \prod_t \pi_{\theta}(y_t | s_t)$ turns the logarithm into a sum and lets us exchange

the order of summation:

$$\begin{aligned} \mathcal{H} &= - \sum_y \pi_{\theta}(y | x) \sum_{t=1}^T \log \pi_{\theta}(y_t | s_t) \\ &= \sum_{t=1}^T \mathbb{E}_{y \sim \pi_{\theta}} [-\log \pi_{\theta}(y_t | s_t)]. \end{aligned} \quad (4)$$

Each summand is still an expectation over whole responses, but the integrand depends on y only through $y_{\leq t}$. Conditioning on the prefix and applying the tower rule replaces the inner expectation with a per-position entropy,

$$\begin{aligned} \mathbb{E}_y [-\log \pi_{\theta}(y_t | s_t)] &= \mathbb{E}_{y_{<t}} \left[\underbrace{\mathbb{E}_{y_t \sim \pi_{\theta}(\cdot | s_t)} [-\log \pi_{\theta}(y_t | s_t)]}_{= H(\pi_{\theta}(\cdot | s_t))} \right], \end{aligned}$$

because the inner expectation is, by definition, the entropy of the conditional distribution at s_t . Collecting the terms back under a single expectation gives

$$\mathcal{H}(\pi_{\theta} | x) = \mathbb{E}_{y \sim \pi_{\theta}} \left[\sum_{t=1}^T H(\pi_{\theta}(\cdot | s_t)) \right]. \quad (5)$$

Equation (5) is an identity, not an approximation. Its left-hand side is what we want to protect; its right-hand side is what instrumentation can measure, since $H(\pi_{\theta}(\cdot | s_t))$ is read directly off the logits at each position.

The entire argument of this paper follows from one feature of Eq. (5) that is easy to pass over: *the measure $\mathbb{E}_{y \sim \pi_{\theta}}$ depends on θ as well*. The right-hand side is not a sum of θ -dependent terms over a fixed set of positions — it is a θ -dependent average of them over a θ -dependent set of trajectories.

4.2 Step 2: the product rule

Write the per-trajectory total

$$f_{\theta}(y) := \sum_{t=1}^T H(\pi_{\theta}(\cdot | s_t(y))), \quad (6)$$

so that Eq. (5) reads $\mathcal{H} = \sum_y \pi_{\theta}(y | x) f_{\theta}(y)$. Both factors of every summand carry θ , so differentiating requires the product rule:

$$\begin{aligned} \nabla_{\theta} \mathcal{H} &= \sum_y [\nabla_{\theta} \pi_{\theta}(y | x)] f_{\theta}(y) \\ &\quad + \sum_y \pi_{\theta}(y | x) \nabla_{\theta} f_{\theta}(y). \end{aligned} \quad (7)$$

The second sum is already an expectation. The first is not, because $\nabla_{\theta}\pi_{\theta}$ is not a probability distribution; the score-function identity $\nabla_{\theta}\pi_{\theta} = \pi_{\theta}\nabla_{\theta}\log\pi_{\theta}$ restores it to one. Substituting and naming the two terms,

$$\nabla_{\theta}\mathcal{H} = \underbrace{\mathbb{E}[\nabla_{\theta}f_{\theta}(y)]}_{\text{(L) local}} + \underbrace{\mathbb{E}[f_{\theta}(y)\nabla_{\theta}\log\pi_{\theta}(y|x)]}_{\text{(V) visitation}}. \quad (8)$$

Equations (3)–(8) contain no approximation of any kind. The split into two channels is forced by the product rule, not chosen.

4.3 Reading the two terms

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the subtree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_{\theta}\mathcal{H}$.

4.4 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level.

Collecting the constants of Eq. (1) into $c := \eta \mathbb{I}_{\text{clip}} A / (L \pi_{\text{old}})$ and rewriting in terms of surprisal gives, for the contribution of a single action a ,

$$\Omega \propto c \pi_a (1 - \pi_a) (H - I_a), \quad (9)$$

and $H = \mathbb{E}_{a \sim \pi}[I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the

magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy. Nothing in c depends on the shape of $\pi(\cdot | s)$, so the argument below is unaffected by the clipping indicator or the importance weight.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a , so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability — are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer.

This is the step that closes the argument, and it follows from Hao et al.’s own derivation. Differentiating the conditional entropy with respect to a single logit gives (Hao et al., 2026, Eq. 20)

$$\frac{\partial H(\pi_{\theta} | s)}{\partial z_{s,a}} = -\pi_{\theta}(a | s) (\log \pi_{\theta}(a | s) + H(\pi_{\theta} | s)), \quad (10)$$

and at a uniform distribution $\log \pi_{\theta}(a | s) = -\log k = -H$ makes the bracket vanish for every a . So $\partial H / \partial z_a = 0$ identically: Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^{\delta} + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^{\delta}}{S}, \quad H''(0) = -\frac{k-1}{k^2}, \quad (11)$$

with $H'(\delta)$ of sign $-\text{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4} , roughly two orders of magnitude below that of a peaked distribution under the same perturbation. *As far as local entropy is concerned, uniform branch points are genuinely safe*, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are anti-correlated. The visitation term’s prefactor, derived in Section 7, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates

distribution	H	$P(\text{split})$	$\mathbb{E} \Omega / A $	$\mathbb{E}[1-\pi_a]$
$q=.99$	0.10	0.08	0.092	0.020
$q=.9, \text{ tail}$	1.25	0.57	1.129	0.190
$q=.6, m=3$	1.11	0.98	0.457	0.587
unif. $k=10$	2.30	1.00	0.000	0.900
unif. $k=100$	4.61	1.00	0.000	0.990

Table 1: The local channel and the visitation channel are ordered oppositely. Distribution family: top token has mass q , the remaining $1 - q$ is spread uniformly over m tokens. $P(\text{split})$ is the probability that $n=8$ samples do not all agree. $\mathbb{E}|\Omega|$ is taken over the token actually sampled at that position.

between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

The statement we can therefore make is stronger than “the local term is incomplete”: *the local term is anti-correlated with the term it omits*, and the omission cannot be fixed within the local channel.

5 Estimating the Visitation Channel

5.1 Causal elimination gives a reward-to-go

The visitation term of Eq. (8) is not yet in a usable form: the factor $f_\theta(y)$ multiplying the score is a property of the *whole* trajectory, so a naive estimator would credit every token with entropy that appeared before it. Causality removes those terms exactly.

Expanding both factors — f_θ by Eq. (6) and the score by $\nabla_\theta \log \pi_\theta(y \mid x) = \sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)$ — turns (V) into a double sum:

$$(V) = \sum_t \sum_u \mathbb{E}[H(\pi_\theta(\cdot \mid s_t)) \cdot \nabla_\theta \log \pi_\theta(y_u \mid s_u)]. \quad (12)$$

Consider a term with $u \geq t$. The entropy $H(\pi_\theta(\cdot \mid s_t))$ depends on y only through $y_{<t}$, and $y_{<t} \subseteq y_{<u}$, so it is measurable with respect to $y_{<u}$ and may be pulled out of a conditional expectation given $y_{<u}$. What remains is the expected score at a single position, which vanishes for any distribution:

$$\begin{aligned} & \mathbb{E}[\nabla_\theta \log \pi_\theta(y_u \mid s_u) \mid y_{<u}] \\ &= \sum_a \pi_\theta(a \mid s_u) \frac{\nabla_\theta \pi_\theta(a \mid s_u)}{\pi_\theta(a \mid s_u)} \\ &= \nabla_\theta \sum_a \pi_\theta(a \mid s_u) = \nabla_\theta 1 = 0. \end{aligned} \quad (13)$$

By the tower rule every (t, u) term with $u \geq t$ is therefore *exactly* zero — not small, zero. This is the same identity that licenses a baseline in policy gradients, used here to enforce causality rather than to reduce variance.

Only $u < t$ survives. Swapping the order of summation collects, for each token u , the entropies at all later positions:

$$(V) = \mathbb{E}\left[\sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u) H_{\text{togo}}(s_u \oplus y_u)\right], \quad (14)$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot \mid s_t))$.

We stress that H_{togo} was not posited. Equation (14) is a REINFORCE estimator whose reward is “the conditional entropy remaining after this token”, and H_{togo} is that reward-to-go.

5.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (14) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \quad (15)$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\leq u}$: the rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO’s group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout’s own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

6 Estimating the Reward-to-Go

Equation (15) is exact but not computable as written: H_{togo} sums the conditional entropies over the whole remaining response. Two approximations make it computable, and a third, in Section 7, bounds its influence.

6.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\text{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k H(\pi(\cdot \mid s_{u+k})). \quad (16)$$

The discount serves two purposes: distant terms are further from the token they are credited to and add variance without adding much signal, and the sum stays bounded independently of κ .

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (15) consumes only the *difference* between siblings. Any bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

6.2 A2: one realised continuation

Equation (16) needs the future states s_{u+k} . In RLVR they exist: the update is made after the rollout has finished, every response in the batch carries the continuation it actually took, and the trainer evaluates the policy’s conditional entropy $H(\pi(\cdot \mid s_t))$ at every response position in the same forward pass that produces the log-probabilities. We therefore evaluate Eq. (16) on the sampled continuation,

$$\begin{aligned} \hat{H}_{\text{togo}}(s_u \oplus y_u) \\ = \sum_{k=1}^{\kappa} \gamma_H^k H(\pi(\cdot \mid s_{u+k})), \end{aligned} \quad (17)$$

where the s_{u+k} are the states of the sampled response. This is not an approximation of the estimator but the estimator itself: the reward-to-go of Eq. (14) is a sum over the *same* trajectory whose score it multiplies, and a REINFORCE estimator uses the sampled trajectory’s value. What A2 names is the price of that: $\hat{H}_{\text{togo}}$ is a single draw of the continuation, unbiased for the truncated sum and carrying the variance of one sample. It costs nothing — no parameters, no warm-up, no forward pass beyond the one the trainer already makes — and it needs no alignment shift, because $s_{u+1} = s_u \oplus y_u$ already reads the state after the branch taken.

Two consequences recur below. The value at u depends on the sampled tokens $y_{u+1:u+\kappa-1}$ as well as on $y_{\leq u}$, which changes where A_H can be non-zero (Section 9). And the states come from π_{old} while the entropies are the current policy’s

at those states, the same mismatch B3 records for every term of the loss.

6.3 An alternative: forecasting the future

The variance of one continuation can be traded for a bias by predicting the future instead of reading it. We implement that alternative and run it as an ablation (Section 12.6): parallel heads read the current hidden state h_u and forecast the entropy k steps ahead,

$$H(\pi(\cdot \mid s_{u+k})) \rightarrow H(p_{\text{MTP}}^{(k)}(\cdot \mid h_u)). \quad (18)$$

Head k is a two-layer residual MLP applied to h_u , followed by the policy’s own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy’s own next-token distribution. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

M1: the forecast’s error has a known sign.

Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u , the k -th term of Eq. (16) has expectation $H(Y_{u+k} \mid Y_{u+1:u+k-1}, S_u)$, whereas the head models the *marginal* $p(Y_{u+k} \mid S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} \mid S_u) - \mathbb{E}[H(\pi(\cdot \mid S_{u+k-1}))] \\ = I(Y_{u+k}; Y_{u+1:u+k-1} \mid S_u) \\ \geq 0. \end{aligned} \quad (19)$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k . For $k = 1$ there are no intervening tokens and the excess vanishes.

M2: per-head affine calibration.

Equation (19) predicts a per-head, monotone inflation, so the forecast is corrected by a per-head affine map fitted by least squares against the realised entropies of warm-up rollouts of the base model,

$$\hat{H}_k = a_k H(p_{\text{MTP}}^{(k)}(\cdot \mid h_u)) + b_k, \quad (20)$$

and the forecast $\max(0, \sum_k \gamma_H^k \hat{H}_k)$ replaces Eq. (17). Its hidden state at position u predicts y_u and does not know it, so the value for $H_{\text{togo}}(s_u \oplus y_u)$ is read at position $u + 1$ and shifted back by one. The fitted coefficients are released with the heads.

Which one, and why both are run. The realised value is the reward-to-go the derivation asks for, sampled once; the forecast is an estimate of its expectation given h_u , lower in variance and biased by Eq. (19) until M2 removes it. Nothing in the derivation decides which of those two errors costs more inside a training loop, and a measurement on the base policy (Section 12.7) finds the calibrated forecast close to a scalar multiple of the realised next-step term once both are differenced against siblings. We take the realised value as the method because it is exact where the forecast is approximate and free where the forecast needs a warm-up and a second forward pass, and we run the forecaster at every seed so that the choice is measured rather than assumed.

7 From Gradient to Token Weights

Equation (15) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u , changes the trajectory entropy.

How far one step moves a logit. We reuse the logit-space update of Hao et al. (2026, Eq. 23) rather than rederiving it. With $w_u = \text{clip}(r_u, 1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}})A_u$ the coefficient the policy loss actually applies, Assumption 2 makes a single step move only the sampled token’s logit, $\Delta z_{u,a} = \eta w_u \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi_u} = \eta w_u (1 - \pi_u), \quad (21)$$

which vanishes as $\pi_u \rightarrow 1$: a token that already holds the mass cannot be given more. This is the prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_{\theta} \log \pi_u \cdot \Delta \theta = \Delta \log \pi_u$, substituting into Eq. (15) gives the per-token quantity

$$\text{visit}_u := \widehat{\Delta \log \pi_u} \cdot \text{clip}(A_H(s_u, y_u), -c, +c). \quad (22)$$

A3 is the clip, and it is applied to A_H *before* the product rather than after, so that one token’s influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let estimation error be amplified without bound at tokens with large w_u .

Sign quadrants. Equation (22) defines the collapse channel precisely. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi_u}$ reverses both readings.

8 Combining the Channels

The visitation term could be added to Ω before the weight map, but measured on real rollouts that map destroys the signal: Ω carries a factor $1/\pi_{\text{old}}$, so $\max |\Omega|/\text{median}|\Omega|$ reaches 3×10^4 and the band is spent on one or two outliers while the bulk lands on a point. We therefore apply the correction *after* the map, as a bounded adjustment of the weights the base method has already produced.

Let $\mathcal{S} = \{u : \text{visit}_u \neq 0\}$ be the support — by Eq. (24), the branch points — and let $b = \lambda_{\text{max}} - \lambda_{\text{min}}$ be the width of the weight band. For $u \in \mathcal{S}$,

$$\alpha_u^f = \text{clamp}\left(\alpha_u + \lambda b \tanh(\text{visit}_u/\rho), \lambda_{\text{min}}, \lambda_{\text{max}}\right), \quad (23)$$

with $\rho = \text{rms}\{\text{visit}_v : v \in \mathcal{S}\}$; $\alpha_u^f = \alpha_u$ elsewhere.

Three properties matter. The tanh is odd, monotone and bounded, so it preserves the sign of the visitation term, caps one token’s movement at λb , and does not depend on the absolute scale of visit. The RMS is taken over $\mathcal{S}$ alone rather than over the whole response: normalizing by the whole response would shrink the correction in proportion to how sparse branch points are, which is a property of the sampler and not of the signal. And at $\lambda = 0$ the correction vanishes identically, so the pipeline is bitwise identical to the base method — a regression test enforces this.

Because the map runs first and the correction second, the step size η of Eq. (21) enters only through visit_u/ρ , where it cancels. We fix $\eta = 1$ and sweep λ alone.

The support is part of the treatment. $\mathcal{S}$ is defined by $\text{visit}_u \neq 0$ rather than by a threshold, deliberately: it selects both where the correction lands and, through ρ , how large every correction is, so a tighter test would change what is trained rather than what is logged. One consequence is that $\mathcal{S}$ admits

float32 round-off — A_H is a deviation from a mean over token-identical siblings, and $\sum_j g_j/m$ does not reproduce g_i bitwise, leaving $|A_H| \sim 6 \times 10^{-8}$ where it is interpretively zero. Those residuals are $\sim 10^{-6}$ relative to a real branch score and cannot move a weight, but they are counted in the support statistics of Section 12.12, which should be read as an upper bound on the branch-point fraction.

9 Tree-Structured Rollouts

9.1 Why plain rollouts cannot support A_H

$\hat{H}_{\text{togo}}(s_u \oplus y_u)$ in Eq. (17) reads the states $s_{u+1}, \dots, s_{u+\kappa}$, so it is a deterministic function of the prefix $y_{\leq u+\kappa-1}$, and two rollouts that share that prefix produce identical values. Combined with Eq. (15), whose baseline averages over rollouts sharing $y_{<u}$, this gives a sharp characterization:

$$A_H[i, u] \neq 0 \implies \text{siblings share } y_{<u} \text{ but differ in } y_{u:u+\kappa-1}, \quad (24)$$

that is, u is a branch point of the group or lies within $\kappa - 1$ positions before one. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so the sibling set is the rollout alone almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

The $\kappa - 1$ positions before a branch are specific to the realised estimator, and they are worth naming. There the siblings still emit the same token from the same state, and A_H credits that one token, in each sibling, with a divergence that happens after it; the forecast of Section 6.3, a function of $y_{\leq u}$ alone, is zero there. With $\kappa = 2$ it is one position per branch, weighted by γ_H^2 against γ_H for the branch token itself.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths $d_1 < d_2 < d_3$ with branching factor 2, yielding $n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (24) can fire.

9.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not, and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (24) it counts positions at or just before branch points, and n sequences form a trie with at most $n - 1$ internal branching nodes.

Each branch node contributes at most κ positions, so the fraction of positions carrying a non-zero A_H is bounded by $\kappa(n - 1)/T$ for *any* sampler whatsoever; with $\kappa = 2$ this is the same $2(n - 1)/T$ that bounds the forecast variant, whose one-position alignment shift widens each branch node into two columns. With $n = 8$ and responses of $\sim 10^3$ tokens the ceiling is on the order of a percent. The tree’s role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

9.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \text{sib}(s_u)} A_H(s_u, y_u^{(j)}) = 0. \quad (25)$$

The correction in Eq. (23) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. “Mean entropy at branch points increased” is *not* a prediction of this method and cannot serve as evidence that it works; the corresponding prediction is that the *variance across siblings* widens. A method designed to raise the mean is a different and weaker hypothesis — it does not need A_H ’s value at all, only a support — and we include it below as a control arm.

10 The Approximation Ledger

Table 2 collects every departure from Eq. (8).

The strongest assumption is B2 — that updating one token does not change the conditional distribution at other positions. It is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (5)–(15)	nothing — identities	—	—
A1	horizon κ , discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	one realised continuation	nothing in expectation; adds the variance of a single draw	unbiased	sibling baseline; the forecaster arm trades it for bias
A3	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
M1	realised future $\rightarrow$ MTP marginal	mutual information with intervening tokens, Eq. (19)	over-estimate, monotone in k	M2
M2	per-head affine calibration	non-affine residual of M1	least-squares residual	the realised-value arms
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	<i>inherited from the base method</i>
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A3 are introduced by this work and apply to STEER-V; M1–M2 apply only to the forecaster variant run as an ablation; B1–B4 are shared with the token-level entropy-control method we extend.

a second channel adds a second copy of its error. STEER-V itself adds no approximation with a signed error: A1’s bias is common to siblings and A2 is unbiased. The forecaster’s M1 is signed and monotone, which is what makes the affine correction M2 an appropriate remedy rather than a curve fit.

11 Experimental Setup

11.1 Model, data, and optimization

All runs fine-tune the same 1.5B-parameter mathematics-specialized base model on the same 17k-problem competition-mathematics set, with the same optimizer settings, and differ only in the intervention. Every run trains for 150 optimization steps; with a training batch of 512 prompts that is about 4.4 epochs. Two further backbones carry the headline contrast only (Section 12.10).

Low-rank adaptation. Every arm, the baselines included, trains a LoRA adapter (Hu et al., 2022) of rank 64 and scale $\alpha = 32$ on every linear projection of the attention and MLP blocks; the embedding and the unembedding stay frozen, so the unembedding the forecaster of Section 6.3 reads is the base model’s. The learning rate is $1e-5$, constant, with no warm-up — ten times the rate full fine-tuning of this model would use, the usual ratio for LoRA, fixed before any run and identical across arms. The reason for LoRA is compute, not a property we claim for it: it is what allows every arm three seeds

inside our budget.

Policy optimization. GRPO with $n = 8$ responses per prompt, a maximum response length of 3072 tokens, no entropy bonus and no KL term. The clipped surrogate is asymmetric, $\varepsilon_{\text{low}} = 0.2$ and $\varepsilon_{\text{high}} = 0.28$, with the dual-clip bound at $c = 10$; we state this because “GRPO” with a symmetric ε and a KL term is a different baseline, and because the follow-up arm of Section 12.8 is defined by moving these numbers. Rollouts are sampled at temperature 1.0 without nucleus truncation. Each Qwen2.5-Math-1.5B run trains on a single H100 (80 GB), four runs side by side on one node, with the rollout engine’s CUDA graphs enabled; the backbone runs of Section 12.10 use the whole four-GPU node. Every arm of a table shares its layout, which each released log records.

Validation and checkpoints. Every 10 steps each run is validated on AIME 2024 (30 problems, 32 samples each, temperature 1.0, top- p 0.7). A single such measurement has an across-problem standard error of about $\sim$, which is why no statistic below rests on one validation point. Each run keeps its checkpoints at steps 50, 100 and 150 and at its best AIME24 validation; the last is the primary evaluation point and the best a secondary one (Section 11.4).

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
UNIFORM	.25	tree	support only
PERMUTED	.25	tree	A_H shuffled
STEER-V	.25	tree	A_H , realised H_{togo}
STEER-V (MTP)	.25	tree	A_H , forecast H_{togo}

Table 3: The six arms, each at three seeds. UNIFORM and PERMUTED run the full pipeline with A_H ’s value destroyed in two different ways, so a gain they share is the apparatus’s; STEER-V (MTP) changes only the estimator.

11.2 Method hyperparameters

H_{togo} is truncated at $\kappa = 2$ with $\gamma_H = 0.7$ (Eq. (17)). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, and the sibling-prefix baseline. The weight map is the base method’s exponential map of Eq. (2) at $\lambda_{\min} = 0.7$, the value its authors report, so weights land in $[0.7, 1]$ and can only attenuate; its released script sets 0.8, and Section 12.8 runs that too. Every weighted arm uses the same band. The tree sampler uses a single root, cuts at depths $\{64, 192, 384\}$, and branching factor 2 at each cut, giving $n = 8$ leaves. The forecaster arm uses two heads with the calibration of Eq. (20), fitted once on warm-up rollouts of the base model.

11.3 Six arms

The design isolates two questions. Does the *value* of A_H matter, or only the apparatus that carries it? A comparison between the base method and the full method changes several things at once (the correction and the sampler), so it cannot answer this; we therefore add two arms that keep the entire apparatus — the tree, the support, the weight map — and destroy only the information in A_H . And does the estimator of H_{togo} matter? A sixth arm keeps everything and replaces the realised value with the forecast.

UNIFORM applies the same attenuation to every position in the support, discarding A_H ’s value but keeping its support — the “raise the mean” hypothesis that Section 9.3 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuffles the assignment, so the support, the magnitudes, and Eq. (25) all survive while the pairing between a branch and its score is destroyed. Both read the same realised H_{togo} as STEER-V, so each contrast against STEER-V changes one thing.

11.4 Endpoints, fixed before the first run

The analysis below was committed before any run of this campaign (Appendix C); the seed is the unit of analysis throughout.

Primary. Accuracy on six mathematics benchmarks — AIME24, AIME25, AMC23, MATH500, Minerva and OlympiadBench — at 32 samples per problem, at the last training step of each run. No checkpoint is selected, so AIME24 is as held-out as the other five. The registered reading is *direction consistency*: on how many of the six a contrast favours the treatment, averaged over seeds — four or more is support, three equivocal, two or fewer evidence against — together with the paired across-problem difference on MATH500, OlympiadBench and Minerva, whose sizes support one.

Secondary. The same evaluation at each run’s AIME24-best checkpoint, where AIME24 is in-sample; and AIME24 accuracy averaged over the validation points of steps 40–150 (the *converged window*), which is what the training logs carry at every seed.

Contrasts. STEER-V against GRPO, STEER, UNIFORM, PERMUTED and STEER-V (MTP); STEER against GRPO; UNIFORM against STEER. The contrasts against UNIFORM and PERMUTED change one thing each and carry the claim.

What would count against the claim. (1) STEER-V — UNIFORM is equivocal or against on the primary endpoint, or straddles zero across seeds: the value of A_H is not what helps. (2) The same for STEER-V — PERMUTED: the pairing of a branch with its score is not what helps. (3) The dissociation of Section 12.3 fails. (4) The mean token weight of STEER-V leaves $[0.99, 1]$: the arm is a change of learning rate, not a reweighting.

Metrics. acc@32 is the mean accuracy over 32 samples and maj@32 the accuracy of their majority vote; $\text{uplift} := \text{maj@32} - \text{acc@32}$ measures whether correct answers concentrate while incorrect ones scatter. The logged actor/entropy is a token-mean, i.e. $\mathcal{H}$ divided by the mean response length, and every entropy number below is that quantity.

12 Results

Every number in this section is generated from the released training and evaluation logs of this

arm	n	acc@32	maj@32	uplift	entropy
GRPO	-	$\pm$	$\pm$	-	-
STEER	-	$\pm$	$\pm$	-	-
UNIFORM	-	$\pm$	$\pm$	-	-
PERMUTED	-	$\pm$	$\pm$	-	-
STEER-V	-	$\pm$	$\pm$	-	-
STEER-V (MTP)	-	$\pm$	$\pm$	-	-

Table 4: AIME24 over the converged window (steps 40–150), each arm averaged over the seeds – with the standard error across them. The $\pm$ is over three draws, not a precision. *uplift* is maj@32 – acc@32; entropy is the token-mean quantity of Section 11.

contrast	n	$\Delta\text{ent.}$	t	Δacc	t
<i>the apparatus changes, A_H does not</i>					
STEER – GRPO	-	-	-	-	-
UNIF – STEER	-	-	-	-	-
<i>A_H changes, the apparatus does not</i>					
STEER-V – UNIF	-	-	-	-	-
STEER-V – PERM	-	-	-	-	-

Table 5: Converged aggregate entropy and accuracy, paired within a seed and grouped by what the contrast changes. The registered reading compares the $|t|$ of the two groups on entropy.

campaign by the scripts named in Appendix A; a number no run has produced yet prints as a red dash. No figure is taken from prior work, and the readings below are the ones registered before the first run (Section 11.4). Arms are averaged over the seeds –, the same set for every arm, so a difference of two rows of Table 4 is the paired contrast of Table 6.

12.1 Does entropy collapse under LoRA at all?

A low-rank update constrains how the policy can move, and it could slow the sharpening that collapse describes enough to leave the method nothing to correct. So the first registered measurement is GRPO’s own entropy. Averaged over its – seeds, GRPO’s token-mean entropy goes from – at step 10 to – at step 150, a ratio of –. – of – seeds fall below half their step-10 value, which is the registered definition of collapse, so under this configuration GRPO –. If it does not, the method’s motivation by collapse does not apply to this regime, and the rest of this section is about redistribution under LoRA rather than rescue from collapse.

12.2 Converged window

Table 4 averages each arm over the validation points of steps 40–150.

12.3 Entropy: does the aggregate move with the apparatus or with A_H ?

What the construction predicts. Section 9 bounds the positions at which A_H can be non-zero at about one in a hundred, and Section 9.3 shows that A_H sums to zero over a sibling set, so at the positions it reaches it redistributes weight among siblings rather than adding any. A correction this sparse and this mean-preserving should leave an average over every response token close to where it found it, while the damping apparatus — STEER’s

local weights, and the tree — acts on every token. The registered test is therefore a comparison between two groups of contrasts: those that switch the apparatus should move the aggregate entropy, and those that change only what A_H carries should not.

What the seeds show. Switching the apparatus moves the aggregate by – ($t = -$) when STEER’s damping is turned on and by – ($t = -$) when the tree is added with the damping applied uniformly. Changing only A_H ’s contents moves it by – ($t = -$) when its value is discarded and by – ($t = -$) when its pairing is destroyed. The registered test finds the two groups’ $|t|$ –; “separate” is the predicted pattern and “overlap” meets refutation condition (3). With three seeds the statement is about what is detectable: that the aggregate does not move *much* when A_H changes, not that it is unchanged.

The prediction this does not test. What a sum-zero correction positively predicts is that the spread *across siblings* widens at the positions it reaches. The trainer logs the entropy on $A_H \neq 0$ against the rest — the predicate the correction itself uses — and STEER-V – PERMUTED on that gap is –; it measures a level, not a spread, and a spread across siblings is not logged. We register it as the measurement that is still owed.

12.4 Does the value of A_H matter?

Table 6 gives every registered contrast on the converged window. The two that carry the claim change one thing each. Across the – paired seeds, STEER-V – UNIFORM runs from – to – with – negative, and STEER-V – PERMUTED from – to – with – negative; a contrast with any negative seed straddles zero, which is half of refutation conditions (1) and (2). The other half is decided on the primary endpoint, in the next section. The mean

contrast	n	Δacc	t	Δmaj	t
STEER-V – GRPO	–	–	–	–	–
STEER-V – STEER	–	–	–	–	–
STEER-V – UNIF	–	–	–	–	–
STEER-V – PERM	–	–	–	–	–
STEER-V – MTP	–	–	–	–	–
STEER – GRPO	–	–	–	–	–
UNIF – STEER	–	–	–	–	–

Table 6: Converged-window contrasts paired by seed: each arm is averaged over its window, arms are paired within a seed, and t is the paired t on the n seed-level differences ($\text{df} = n - 1$).

arm	AIME24	AIME25	AMC23	MATH500	Minerva	Olympiad	mean
	–	–	–	–	–	–	–

Table 7: Six-benchmark accuracy (% , 32 samples per problem) at the last training step, averaged over the seeds in the last column. No checkpoint is selected, so every column is held out. The mean is the unweighted mean over the six.

token weight of STEER-V is $-\pm$, against the $[0.99, 1]$ that condition (4) requires.

The contrasts against GRPO and STEER are registered too, but they change several things at once — the sampler, the correction, and against GRPO the damping — so a difference there names no cause.

12.5 Six benchmarks at the last step

Each run is evaluated at step 150 on six benchmarks at 32 samples per problem (Table 7). An evaluation enters the table only if its log agrees with itself: the trainer prints each benchmark as accuracy and as the mean of a ± 1 reward, so accuracy must equal half of one plus the reward, and the release checks this for every log (`scripts/check_eval_logs.py`). It is a test of coherence, not of provenance. Not yet in the table: $-\pm$.

Direction consistency. STEER-V leads UNIFORM on $-\pm$ of six benchmarks, which the registered thresholds call $-\pm$, over the seeds the two arms share ($-\pm$); it leads PERMUTED on $-\pm$ of six, $-\pm$, over $-\pm$. Together with the straddle counts above, these decide refutation conditions (1) and (2).

The paired error bar. The two arms of a contrast answer the same questions, so per-problem difficulty cancels between them and what matters is the standard error of the *paired* difference. On MATH500, OlympiadBench and Minerva — $-\pm$ problems, over $-\pm$ seeds — STEER-V – UNIFORM is $-\pm$ ($t = -$, unpaired error $-\pm$) and STEER-V –

PERMUTED is $-\pm$ ($t = -$). Two cautions travel with these. Pooling problems across seeds treats a (seed, problem) pair as one observation, which ignores clustering by seed; and GSM8K is left out of the pool because it sits near saturation, where a paired difference measures the ceiling.

At the AIME24-best checkpoint. The secondary endpoint repeats the evaluation at each run’s best validation step, where AIME24 is in-sample and the other five are not (Appendix D). There STEER-V leads UNIFORM on $-\pm$ of six ($-\pm$) and on $-\pm$ of the $-\pm$ held out; the paired difference is $-\pm$.

12.6 Does the estimator matter?

STEER-V (MTP) keeps everything and replaces the realised H_{togo} with the calibrated forecast of Section 6.3. On the converged window, STEER-V – MTP is $-\pm$ ($t = -$, $n = -$). On the primary endpoint the paired difference is $-\pm$ ($t = -$), which the rule fixed in advance reads as $-\pm$. The forecaster costs its forward pass: a step takes $-\pm$ for STEER-V and $-\pm$ for the MTP variant on the same hardware. Whichever way the accuracy comparison falls, that is the price the forecast is measured against.

12.7 Is the forecast a local term?

A measurement on the base policy, made before any training, explains why the two estimators could behave alike. We apply the *same* operator — the same sibling baseline, mask and grouping — to the calibrated forecast and to the policy’s own next-token entropy read at the state after the branch token. Over 464 branch points from 4 validation prompts, the two advantages correlate at $r = +0.917$ (Spearman $+0.937$), the local quantity explains $R^2 = 0.840$ of the forecast’s, they agree in sign 0.921 of the time — the sign is what the weights are monotone in — and the magnitudes stand in the ratio 0.741, close to the first head’s calibrated weight times γ_H . Once differenced against siblings, the calibrated forecast is very nearly a scalar multiple of the first term of Eq. (17), and Eq. (23) divides scale out through ρ . That is a reason to expect the realised estimator to lose little, not a proof of it; Section 12.6 is the proof either way. The measurement is on a frozen policy and independently sampled rollouts, so its sibling support sits early in the response rather than at the scheduled cuts.

arm	what changes	acc@32	maj@32	Δ vs STEER-V
<i>the coefficient</i>				
$\lambda = 0.1$	weaker	-	-	-
$\lambda = 0.5$	stronger	-	-	-
<i>the sampler, with the correction off</i>				
$\lambda = 0$, tree	tree only	-	-	-
<i>the base method at its released damping</i>				
STEER, $\lambda_{\min}=0.8$	weaker	-	-	-
<i>relaxed ratio clipping ($\epsilon_{hi}=5, \epsilon_{lo}=0.99$)</i>				
STEER-V	—	-	-	-
STEER	—	-	-	-
<i>other advantage estimators</i>				
RLOO, STEER-V	—	-	-	-
RLOO, STEER	—	-	-	-
OPO, STEER-V	—	-	-	-
OPO, STEER	—	-	-	-

Table 8: Follow-up ablations, converged window, one seed per row and no error bar. The last column is the row minus STEER-V at the same seed. Each block isolates one element of the design.

12.8 How much of the design is load-bearing

Table 8 varies one element at a time, at one seed per row. Each row is read against STEER-V *at the same seed* (the last column), not against Table 4’s across-seed mean; with $n = 1$ these rows describe a mechanism and license no effect size.

The coefficient. λ was not tuned on these runs; the sweep says whether the result rests on a knife edge. A flat response across 0.1–0.5 says the method is insensitive to it, a peak at 0.25 that it is a tuned quantity.

The sampler. Tree rollouts make a group share prefixes, which changes the GRPO advantage estimate itself, apart from any damping. UNIFORM and PERMUTED run at $\lambda = .25$ and cannot separate the sampler from the correction; the $\lambda = 0$ tree arm switches the correction off and keeps the sampler. Read against STEER, which has neither, it is the sampler’s own contribution; read against STEER-V, what remains is the correction’s.

The released damping. Our weighted arms use the $\lambda_{\min} = 0.7$ the base method’s authors report; their released script sets 0.8. This row is stock STEER at 0.8, so a reader can see the base method at both.

Clipping and the advantage estimator. Relaxed clipping is where entropy interventions are expected to matter most, because ratio clipping otherwise removes much of the update a token weight would have scaled. The RLOO and OPO rows test whether the construction depends on GRPO: the

arm	steps	hours	acc@32
STEER-V	150	-	-
GRPO, step-matched	150	-	-
GRPO, compute-matched	-	-	-

Table 9: Equal steps against equal compute, on one hardware configuration.

	GRPO	STEER	STEER-V	Δ GRPO	Δ STEER
1.5B Qwen	-	-	-	-	-
7B Qwen	-	-	-	-	-
3B Llama	-	-	-	-	-

Table 10: The headline contrasts on three backbones, converged window of each backbone’s own validation set (AIME24, AIME24, MATH500). The 1.5B row is the across-seed mean; the other two are seed 1. Only the sign of the two right-hand columns is read.

sibling-prefix baseline depends only on the group sharing a prefix, which those estimators also provide. Each is reported beside its own base-method control.

12.9 Compute-matched comparison

An equal-step comparison hands the more expensive arm more wall clock. We therefore integrate GRPO’s measured per-step time and read its accuracy at the last validation whose cumulative time fits inside what STEER-V spent reaching step 150, rather than assuming a cost ratio (Table 9). When GRPO is the cheaper arm, the matched point lies past its last step and the comparison is censored (– here); we report that rather than extrapolate.

12.10 Does the sign survive a change of backbone?

On two further backbones we run GRPO, the base method and STEER-V at one seed and ask whether the *sign* of the headline contrasts holds. The 7B model shares the 1.5B’s family, so a difference between those rows is about scale; the 3B Llama model shares none of it, so a difference there is about pre-training family. Llama is the instruction-tuned checkpoint and validates on MATH500 rather than AIME24: a model near zero on AIME24 can separate neither arms nor checkpoints there. Absolute accuracies are therefore not comparable across rows, and nothing is averaged down a column.

12.11 Seeds and spread

Every statistic above collapses a run to one number over its converged window, pairs arms within a seed, and tests the seed-level differences with $df =$

$n - 1$. Pairing within a seed cancels whatever a seed carries for all arms, the data order first of all. What it does not cancel is run-to-run spread, and that is the number to read the contrasts against: GRPO’s converged-window accuracy moves from $-$ to $-$ across its seeds, STEER-V’s from $-$ to $-$. Rollout sampling is not seeded across processes, so re-running a configuration is a second draw rather than a restoration.

12.12 Mechanism diagnostics and cost

Section 9 bounds the fraction of positions that can carry a non-zero A_H at $\kappa(n - 1)/T$ for any sampler. STEER-V measures $-$ of positions with $A_H \neq 0$, at a mean response length of $-$ tokens; the count admits float32 round-off at positions where siblings are token-identical, so it is an upper bound on the branch-point fraction. One optimization step takes $-$ s for GRPO, $-$ for STEER, $-$ for UNIFORM, $-$ for PERMUTED, $-$ for STEER-V and $-$ for the MTP variant. The three tree arms that read the realised value differ from one another only by a subtraction over a mask, so a gap between them and STEER is the tree sampler’s cost, and the MTP variant adds its forward pass on top.

13 Conclusion

The gradient of trajectory entropy has two terms, and the token-level entropy controls now in use estimate one of them. The omission is structural: the estimated term vanishes exactly where the omitted term is largest, and it vanishes for a correct reason, so the gap cannot be closed by improving the estimator. The omitted term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy; in RLVR that reward-to-go can be read off the rollout the update is already made from, and a tree-structured rollout supplies the sibling baseline that makes it estimable at all. The resulting correction is mean-preserving by construction: it does not slow entropy collapse, it redirects it.

The experiment is built to test those consequences rather than a gain. It separates the correction’s value from its apparatus with two controls that keep the apparatus and destroy the value; it tests the dissociation the construction predicts as a comparison between groups of contrasts; it runs the learned forecaster beside the realised estimator; and it first asks whether entropy collapses under LoRA at all. What each of those returned is in Section 12, read by rules fixed before the first run.

Limitations

LoRA only. Every run trains a rank-64 LoRA adapter. We chose it for compute, every arm uses the identical configuration, and the claims are comparisons between methods under it. Whether they carry over to full fine-tuning is not claimed. Full fine-tuning pilots of an earlier version of this method exist; they are not pooled with anything here and no statement in this paper rests on them.

Three seeds, one benchmark family in training. Every core arm has three seeds, which bounds the effect sizes rather than the structural test: the dissociation is a comparison between two groups of contrasts, and no single row carries it. Validation during training is AIME 2024, 30 problems; the six-benchmark evaluation at the last step is what speaks to other questions, and its paired error pools problems across seeds, which ignores clustering by seed.

Backbones. The full design runs on one 1.5B model. Two further backbones carry the three-way contrast at one seed each, which establishes a sign and not a size; they stop at 7B, and the Llama row differs from the others in validation set and in being instruction-tuned, each of which is why no accuracy is compared across backbones.

The estimator. STEER-V reads one realised continuation, which is unbiased and noisy; the forecaster trades that variance for a signed bias that a calibration must remove. Which error costs more inside a training loop is measured here at three seeds (Section 12.6), not settled in general, and a forecaster whose error did not grow with k would test a longer horizon.

The support is small by construction. At most $\kappa(n - 1)/T$ of positions carry a non-zero A_H — on the order of a percent here. Whether the effect scales with n , or with branch points chosen adaptively rather than at fixed depths, is untested; and the widening of the spread across siblings, which is what a sum-zero correction positively predicts, is not yet measured.

An assumption with no remedy. Approximation B2 — that a token’s update does not change the conditional distribution at other positions — is false in a network with shared parameters. It is inherited from the base method, but our construction adds a second estimator that relies on it.

The damping strength. All weighted arms use $\lambda_{\min} = 0.7$, the value the base method’s authors report; their released script uses 0.8, which we report as a follow-up. The value cancels in every contrast between two weighted arms, including the two that carry the claim.

Changes to the registration. Three changes were made to the registered plan after it was first committed and before any run of this campaign: the training length (to 150 steps) and the compute-matched control, the replacement of one refutation condition by the dissociation test, and the addition of the backbone rows. Each is dated in the released registration.

Ethics Statement

This work studies an optimization-time reweighting for reinforcement learning on mathematics problems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation sets are public.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are not specific to mathematics, and the same mechanism would apply to a reward encoding any objective, including a harmful one; the method inherits, and does not mitigate, whatever the reward encodes. Its computational cost over the base method is the tree sampler, which Section 12.12 quantifies.

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A Implementation Notes

Sibling mask. For a group of G rollouts of length T we form $\text{eq}[i, j, t] = \mathbf{1}[y_t^{(i)} = y_t^{(j)}]$ and take a cumulative product along t , so that $\text{prefixeq}[i, j, t]$ is true exactly when the two rollouts agree on $y_{\leq t}$. Shifting it one position right gives agreement on $y_{< t}$, which is the sibling relation Eq. (15) requires. Padded positions are treated as disagreement, so a length difference counts as a divergence.

The realised reward-to-go. The trainer computes the policy’s entropy at every response position in the pass that produces the log-probabilities; $\hat{H}_{\text{togo}}$ is the discounted sum of the next κ of them, zero past the end of a response, and needs no shift (Section 6). The forecaster variant materializes no head logits: each head’s hidden states are processed in chunks of 4096 rows, mapped through the shared unembedding and reduced to an entropy immediately, and its value is read at $t + 1$ and moved to t .

LoRA in the tree sampler. The tree calls the inference engine once per stage, each with its own number of prompts; every call carries the current adapter, and a run stops if the adapter is missing (which would silently sample from the base model) or if more than one is registered.

Checkpoints and evaluation. A run keeps steps 50, 100 and 150 and its best validation step; a superseded best is deleted only after the new checkpoint is written. For evaluation the adapter is merged into the base model after checking that every adapter tensor reached the model unchanged; an adapter

whose B matrices are all zero is refused, since evaluating it would evaluate the base model.

Equivalence test. A unit test asserts that with $\lambda = 0$ the token weights are bitwise identical to the base method’s on random inputs.

B Closed Forms for Section 4.4

Perturbing one logit of a k -way uniform distribution by δ and writing $S := e^\delta + k - 1$ gives $\pi_1 = e^\delta/S$ and $\pi_j = 1/S$ for $j \neq 1$, so

$$H(\delta) = \log S - \frac{\delta e^\delta}{S},$$

$$H'(\delta) = -\frac{(k-1)\delta e^\delta}{S^2},$$

whence $H'(0) = 0$ and $H''(0) = -(k-1)/k^2$. The sign of H' is $-\text{sign}(\delta)$, so any perturbation lowers the entropy, but only at second order. Table 11 contrasts this with a peaked distribution, where the first-order term does not vanish.

distribution	H_0	Ω	ΔH at $\delta=.1$
uniform, $k=10$	2.303	0	-4.7×10^{-4}
uniform, $k=50$	3.912	0	-1.0×10^{-4}
top .5 + 9 uniform	1.792	-0.549	-5.6×10^{-2}
top .9 + 9 uniform	0.545	-0.396	-3.8×10^{-2}

Table 11: First-order prediction versus actual entropy change. At a uniform branch point the first-order term is exactly zero and the realized damage is two orders of magnitude below that of a peaked state under the same perturbation.

C Pre-Registered Analysis

The analysis plan was committed to the artifact repository before any run of this campaign; the file is docs/preregistration_lora.md in the release, and what follows summarizes it.

Setting. As in Section 11: one base model, the same data, LoRA of rank 64 at learning rate $1e-5$ for every arm, 150 steps, token weights in $[0.7, 1]$, three seeds for the six core arms and one for the follow-ups and backbones.

Endpoints. Primary: six-benchmark accuracy at 32 samples per problem at the last step, read by direction consistency (four or more of six support, three equivocal, two or fewer against) and by the paired across-problem difference. Secondary: the same at the AIME24-best checkpoint, and AIME24 over the converged window of steps 40–150, paired within seed.

Refutation conditions. The four of Section 11.4, stated in advance so they cannot be renegotiated afterwards.

The estimator. The realised-value method was chosen for cost and for removing the forecaster’s warm-up and calibration. In the one pilot that compared the two estimators at a single seed, the forecast led; so the forecaster runs at every seed, and its contrast is read by a rule fixed in advance: within two standard errors the two are reported as not distinguished, not as equivalent.

Collapse under LoRA. GRPO’s entropy at step 10 and at the last step, per seed; collapse is the last value below half the first in every seed. If GRPO does not collapse, the paper says so and does not motivate the method by collapse under LoRA.

Changes before the first run. Recorded with dates in the file: the training length and the compute-matched control; one refutation condition, first copied from the plan of an earlier campaign where its own manuscript had retired it as the wrong instrument, replaced by the dissociation test; and the backbone rows.

D Six Benchmarks at the AIME24-Best Checkpoint

arm	AIME24 [†]	AIME25	AMC23	MATH500	Minerva	Olympiad	mean
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Table 12: The secondary endpoint: each run’s AIME24-best checkpoint. [†]: AIME24 is the set the checkpoint was selected on, so that column is in-sample.